D9 / TESTING ARTIFACTS

Test plan and executed cases

Scope and method

The production project was built in Release. A separate compiled C++ harness links the active application sources and uses an isolated SQLite file and settings profile. It exercises database/rule operations and selected UI handlers, and captures actual Qt widgets. Independent Python sqlite3 checks run against the supplied schema and sample dataset.

Area	Coverage
UI workflow	Welcome/selection/calendar/form/admin, validation feedback and access control
Persistence	Create, read/filter, update, delete and normalized slot rows
Business rules	Active slot conflicts, rejection release, reapproval conflicts and key collisions
Data integrity	Foreign keys, legal statuses, slot composite key and transaction rollback
Error handling	Bad inputs, invalid statuses, occupied slots and wrong passcodes

Executed results

10 of 10 integration test groups passed with 69 assertions and 0 failed assertions. All twenty independent SQL checks passed. A separate UI update/delete continuation passed nineteen additional assertions, recorded in ui_crud_results.json. A handled invalid input counts as a passing test when the application rejects it correctly.

Evidence and test isolation

runtime_results.json includes every expected/observed assertion. SQL verification results record executed failures and integrity checks. Screenshots and logs accompany this plan. Fictional data prevents modifying real student records.

Prepared 07 October 2026. Automated execution time and exact environment are recorded in the machine-readable log.

ACTUAL RESULTS

Executed test cases 1-4

T01 - Schema creation and workshop seed data

Input and actual evidence	Result
Input: Fresh isolated SQLite file; createTable(). Expected: Fresh SQLite schema creates successfully; Schema version is 4 Observed: Operation succeeded; 4	PASS 4 assertions

T02 - Booking create and joined-data round trip

Input and actual evidence	Result
Input: DEMO001, demo001@example.com, Computer, future date, first slot, key 4101. Expected: Valid booking is saved; Saved booking joins student facts and preserves project/supervisor/time Observed: Operation succeeded; id=1; name=Demo Student DEMO001; email=demo001@example.com; supervisor=Dr. Demo Supervisor; project=Sensor calibration study; slot=8.30 am - 10.30 am	PASS 2 assertions

T03 - Active-slot conflict and duplicate-key protection

Input and actual evidence	Result
Input: Second active record on the same workshop/date/slot; separate record reusing key 4101. Expected: Overlapping active booking is rejected; Conflict explains unavailable slot Observed: The selected time slot is already allocated. Choose another time slot.; The selected time slot is already allocated. Choose another time slot.	PASS 4 assertions

T04 - Validation and allocation lifecycle rules

Input and actual evidence	Result
Input: Malformed email, blank name, past date; new record and student/admin update rules. Expected: Database rejects malformed email; UI rule identifies Email field Observed: Enter a valid email address.; field=4	PASS 8 assertions

Concise observations shown here. runtime_results.json retains each assertion and its complete expected/observed values, including exact input values.

ACTUAL RESULTS

Executed test cases 5-8

T05 - Booking update replaces slots and preserves protected facts

Input and actual evidence	Result
Input: Existing booking: updated name/project, two replacement slots, existing key 4101 and admin remark. Expected: Admin remark saves; Valid update saves Observed: Operation succeeded; Operation succeeded	PASS 6 assertions

T06 - Reject releases capacity and conflicting reapproval is rolled back

Input and actual evidence	Result
Input: Reject first record; reuse its released slot; reapprove original; try unknown status. Expected: Rejected status canonicalizes and saves; Rejected-only date is absent from allocated dates Observed: Operation succeeded; allocated dates=0	PASS 6 assertions

T07 - Date and category filtering with deterministic ordering

Input and actual evidence	Result
Input: Computer bookings across two dates plus one Electronics booking. Expected: Independent date and category bookings save; Date/category filter yields two rows ordered by newest ID Observed: Operation succeeded; rows=2; keys=4105,4101	PASS 4 assertions

T08 - Delete cascades slot rows and missing records fail clearly

Input and actual evidence	Result
Input: Existing booking ID with slots; repeat delete and remark update for missing ID. Expected: Existing booking is deleted; Foreign key cascade removes all child slots Observed: Operation succeeded; 0	PASS 5 assertions

Concise observations shown here. runtime_results.json retains each assertion and its complete expected/observed values, including exact input values.

ACTUAL RESULTS

Executed test cases 9-10

T09 - Fictional demo dataset for populated UI evidence

Input and actual evidence	Result
Input: Three fictional UI records with Approve, Pending and Reject. Expected: Three fictional demo bookings seed successfully; Fictional admin remark saves Observed: Operation succeeded; Operation succeeded	PASS 3 assertions

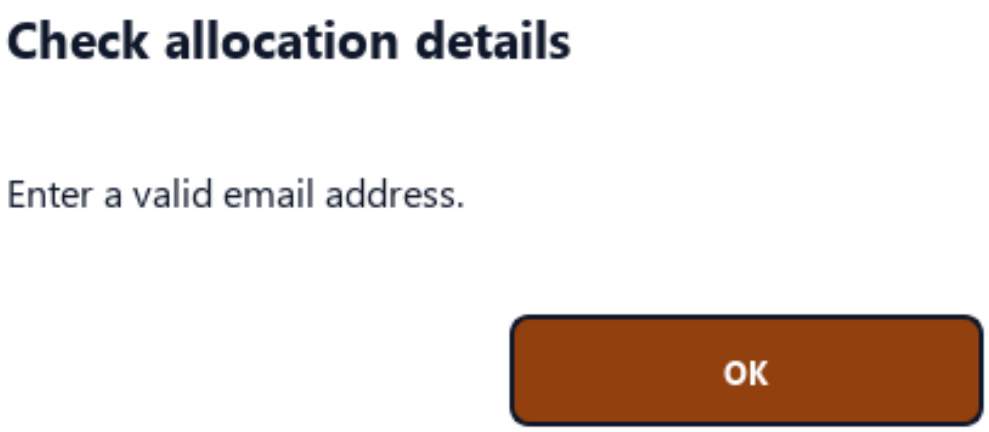
T10 - Implemented UI workflow, validation, save, admin authentication and filters

Input and actual evidence	Result
Input: Computer flow with invalid then valid email; selected free slot; wrong then correct lab passcode; date filter. Expected: Application starts on Welcome; Live QWidget capture saved: 01_welcome.png Observed: page=0; C:/Users/rashm/Desktop/SYNCLAB - Chatgpt/tmp/submission_work/screenshots/01_welcome.png	PASS 27 assertions

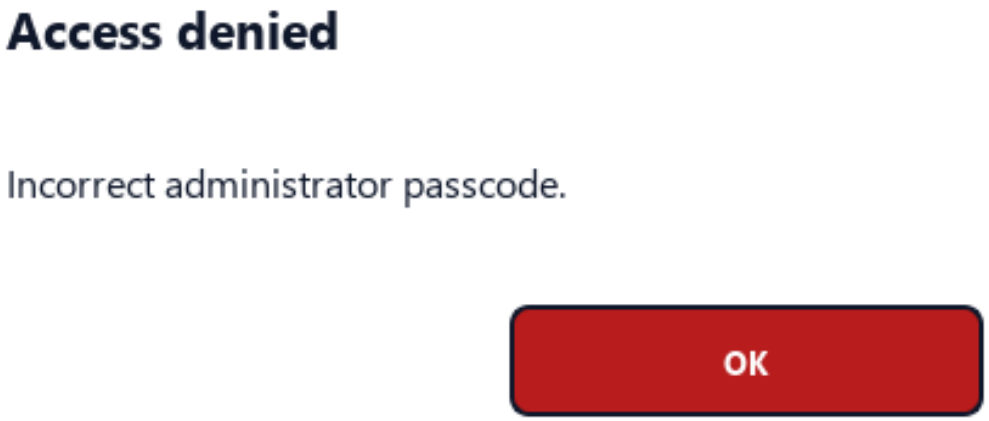
Concise observations shown here. runtime_results.json retains each assertion and its complete expected/observed values, including exact input values.

TESTING INTERPRETATION

Error evidence and remaining checks



Handled case: invalid form input displays a prompt and leaves the database unchanged.



Handled case: wrong workshop passcode blocks administrator access.

Manual acceptance checks to complete at the demo

Confirm layout on the assessment display, email-draft opening with the installed email handler, backup/restore using a closed application, and final team participation. These are planned acceptance checks and are not reported as executed passes.

No performance threshold, broad usability study, accessibility audit or email-delivery success is claimed by these results.